KW 8: Introduction & National Accounts

Chapters 1 & 2

Blanchard - Macroeconomics, 7th Global Edition

Study Notes

CHAPTER 1: A TOUR OF THE WORLD

1-1 The Crisis

Pre-Crisis Period (2000-2007)

- From 2000 to 2007, the world economy experienced a sustained expansion
- Annual average world output growth was 4.5%
- Advanced economies (~30 richest countries) grew at 2.7% per year
- Other economies (~150 countries) grew even faster at 6.6% per year

Origins of the Crisis (2007-2008)

- In 2007, signs emerged that the U.S. expansion was ending
- U.S. housing prices, which had doubled since 2000, started declining
- Two camps formed:

> **Optimists**: Lower housing prices would reduce construction and consumer spending, but the Fed could lower interest rates to stimulate demand and avoid recession

> **Pessimists**: The decrease in interest rates might not be enough; the U.S. might go through a short recession

- The pessimists turned out to be not pessimistic enough

How Housing Became a Financial Crisis

- Many mortgages given during the expansion were of poor quality
- Borrowers had taken too large loans and could not make monthly payments
- With declining housing prices, mortgage values often exceeded house prices, giving incentive to default
- Banks had bundled mortgages into new securities (securitization), sold them to other banks, repackaged into further new securities
- Banks held highly complex securities whose value was nearly impossible to assess
- Banks became reluctant to lend to each other, fearing counterparties might not be able to repay

Figure 1-1: Output Growth Rates

- Shows world output growth rates from 2000-2016
- Advanced economies experienced sharp decline in 2008-2009 (negative growth)
- Other economies slowed but maintained positive growth
- Recovery was slow and uneven across regions

The Crisis Unfolds (2008-2009)

- September 2008: Lehman Brothers went bankrupt - marked the worst phase of the crisis
- Most major banks were highly leveraged and at risk
- Stock markets crashed worldwide (U.S. stock index fell ~30%)
- U.S. output growth was -3.0% in 2009

- Advanced economies collectively contracted by -3.4% in 2009
- World trade collapsed, spreading the crisis to countries that had not experienced housing booms

Policy Response and Recovery

- Central banks cut interest rates to near zero
- Governments implemented fiscal stimulus (increased spending, reduced taxes)
- These measures prevented a repeat of the Great Depression
- Recovery has been slow - called the 'Great Recession'
- By 2010, positive growth returned but remained sluggish in advanced economies

1-2 The United States

The U.S. is the world's largest economy, accounting for about 22% of world output. Understanding U.S. macroeconomic performance is essential.

Key U.S. Macroeconomic Data (Table 1-1)

	2000-07	2008-10	2011-13	2014-16
Output growth %	2.7	-0.3	2.0	2.2
Unempl. rate %	5.0	7.5	8.3	5.3
Inflation rate %	2.5	1.6	1.8	1.1
Interest rate %	3.6	1.6	0.1	0.1

- Output growth averaged 2.7% from 2000-2007, dropped to -0.3% during 2008-10, recovered to about 2% since 2011
- Unemployment rate jumped from 5% pre-crisis to 7.5% during crisis, peaked at 10% in October 2009, slowly came down to around 5% by 2016
- Inflation remained low and stable (under 2.5%), even during the crisis
- The Fed cut the policy interest rate (federal funds rate) from about 3.6% to essentially 0% by 2009

Low Interest Rates and the Zero Lower Bound

- The federal funds rate reached 0% in late 2008 and stayed near zero until 2015
- This is the **zero lower bound** (ZLB) problem - interest rates cannot go below zero (or very slightly negative)
- At the ZLB, conventional monetary policy becomes ineffective
- The Fed turned to **unconventional monetary policy**: buying long-term bonds (quantitative easing/QE) to reduce long-term interest rates

How Worrisome Is Low Productivity Growth?

- U.S. productivity growth has slowed since mid-2000s
- Average annual productivity growth: 2.2% from 1996-2006, only 0.6% from 2006-2016
- This is a major concern because productivity growth is the main driver of long-run improvements in living standards
- Debate exists over whether this is temporary or a new normal:
    - > **Pessimists** (e.g., Robert Gordon): major innovations are behind us, growth will remain low
    - > **Optimists**: new technologies (AI, robotics) will eventually boost productivity; there is a lag between invention and productivity gains

1-3 The Euro Area

The euro area comprises the countries of the EU that have adopted the euro as their common currency. As of 2016, 19 countries, with a combined population of 340 million and combined output roughly equal to that of the United States.

Key Euro Area Macroeconomic Data (Table 1-2)

	2000-07	2008-10	2011-13	2014-16
Output growth %	2.2	-0.5	0.3	1.7
Unempl. rate %	8.4	8.9	10.8	10.8
Inflation rate %	2.2	1.5	1.8	0.4
Interest rate %	3.1	2.1	0.4	0.0

- Pre-crisis growth was lower than the U.S. (2.2% vs 2.7%)
- The crisis hit the Euro area harder and longer - growth was only 0.3% during 2011-2013
- The Euro area experienced a **double-dip recession** - second downturn in 2011-12
- Unemployment remained very high at 10.8% even during 2014-16
- The ECB (European Central Bank) also cut rates to near zero
- Inflation dropped to very low levels (0.4%), raising concerns about deflation

Can European Unemployment Be Reduced?

- European unemployment has been structurally higher than U.S. unemployment for decades
- Some argue this reflects rigid labor markets (strong worker protections, high minimum wages, generous unemployment benefits)
- Others point to insufficient demand (the demand-side explanation)
- Reality: both supply-side (structural) and demand-side factors play a role
- Different Euro area countries have very different unemployment rates:
  > Germany: around 4% (well below the Euro average)
  > Spain and Greece: above 20% during the crisis

What Has the Euro Done for Its Members?

- The euro was introduced in 1999 (notes and coins in 2002)
- **Benefits**: elimination of exchange rate uncertainty, lower transaction costs for trade, potentially lower interest rates for some countries
- **Costs**: loss of independent monetary policy - each country cannot set its own interest rate
- When a country-specific shock hits, it cannot devalue its currency or lower its own interest rate
- The crisis exposed tensions: countries like Greece, Ireland, Portugal, Spain faced sovereign debt crises
- Debate continues about whether the euro area is an **optimal currency area** - whether the benefits outweigh the costs of sharing a single monetary policy

1-4 China

China's rapid economic growth over the past 40 years has transformed it into one of the world's largest economies.

Key China Macroeconomic Data (Table 1-3)

	2000-07	2008-10	2011-13	2014-16

Output growth %	10.5	9.9	8.2	6.8
Inflation rate %	1.7	3.0	3.1	1.7

- China's growth has been extraordinary: averaging 10.5% from 2000-07
- Growth has slowed from 10.5% to 6.8% - but this is still very high by world standards
- China's GDP in purchasing power parity (PPP) terms is now larger than the U.S.
- However, per capita income is still about 1/4 of U.S. levels (PPP-adjusted)
- At market exchange rates, China's GDP is about 60% of U.S. GDP

Key Issues for China

- **Rebalancing growth**: China's growth has been driven by investment (45% of GDP) and exports; needs to shift toward consumption
- **Shadow banking**: Large financial sector outside traditional banking, with less regulation and potentially risky lending
- **Exchange rate management**: China has maintained a partly managed exchange rate, keeping the yuan relatively weak to boost exports
- **Inequality**: Growing income inequality between urban and rural areas, and between coastal and inland regions
- **Environmental costs**: Rapid industrialization has caused severe pollution and environmental degradation

1-5 Looking Ahead

The book is organized around a core set of chapters that provide the fundamental framework for thinking about macroeconomics, with extensions that build on the core.

- The core covers: the goods market, financial markets, the IS-LM model, the labor market, the Phillips curve, and the IS-LM-PC model
- Extensions cover: expectations, the open economy, and policy
- Each extension enriches the basic model to address more realistic and complex questions

Appendix: Where to Find the Numbers

- Key data sources for macroeconomic statistics:
  > **National sources**: Bureau of Economic Analysis (BEA) for U.S. GDP data, Bureau of Labor Statistics (BLS) for employment/inflation data
  > **International organizations**: IMF (World Economic Outlook), OECD, World Bank
  > **Central banks**: Federal Reserve (FRED database), ECB
- The FRED database (Federal Reserve Bank of St. Louis) is particularly useful: free, comprehensive, with easy-to-use graphing tools

CHAPTER 2: A TOUR OF THE BOOK

2-1 Aggregate Output

GDP is the single most important measure of the overall level of economic activity.

GDP: Production and Income

- There are three equivalent ways to think about GDP:

> (1) **From the production side**: GDP is the value of the final goods and services produced in the economy during a given period

> (2) **From the income side**: GDP is the sum of incomes in the economy during a given period

> (3) **From the expenditure side**: GDP is the value of final goods and services purchased during a given period

Key Definitions

Final vs. Intermediate Goods
Final goods: goods destined for final consumption.
Intermediate goods: used in the production of final goods.
GDP counts only final goods to avoid double-counting.
Value added by a firm = value of production - value of intermediate goods used.

- **GDP = sum of value added by all firms in the economy** (equivalently, the value of all final goods)
- Why production equals income: A firm's value added goes to pay workers (wages) or remains as profit to the firm. So the value added is distributed as income.
- Hence: **aggregate production = aggregate income**

Nominal and Real GDP

- **Nominal GDP** = sum of quantities of final goods x their current prices. Also called GDP in current dollars or GDP at current prices.
- Nominal GDP can increase either because production increases or because prices increase
- **Real GDP** = sum of quantities of final goods x constant prices (base year prices). Also called GDP in constant dollars, GDP adjusted for inflation, or GDP in terms of goods.
- Real GDP is constructed using a reference year's prices, so changes in real GDP reflect only changes in quantities, not prices

Nominal vs Real GDP - Example
Economy produces only cars:
Year 1: 10 cars at $20,000 each -> Nominal GDP = $200,000
Year 2: 12 cars at $22,000 each -> Nominal GDP = $264,000
Real GDP (Year 1 prices): Year 2 = 12 x $20,000 = $240,000
Nominal GDP growth: 32%. Real GDP growth: 20%.

- The Bureau of Economic Analysis (BEA) constructs real GDP using 2009 as the reference year
- U.S. real GDP in 2014: $16.2 trillion (2009 dollars). Nominal GDP in 2014: $17.3 trillion

- **GDP growth rate**: the rate of change of real GDP. Positive = expansion. Negative = recession.

GDP: Level versus Growth Rate

- Both the level and the growth rate of real GDP are important:
  > **Level of real GDP** (relative to population): determines the standard of living
  > **Growth rate of real GDP**: determines whether the standard of living is rising or falling
- An economy can have a high level of GDP but low growth (e.g., Japan), or a lower level but high growth (e.g., China)

Expansions and Recessions
Expansion: period of positive GDP growth.
Recession: period of negative GDP growth.
U.S. convention (NBER): A recession requires at least two consecutive quarters of negative GDP growth, plus a significant decline in activity.

2-2 The Unemployment Rate

Definitions

- **Employment (N)**: the number of people who have a job
- **Unemployment (U)**: the number of people who do not have a job but are looking for one
- **Labor force (L)**: employment + unemployment:

$$L = N + U$$

- **Unemployment rate (u)**: the ratio of unemployed to the labor force:

$$u = \frac{U}{L}$$

- **Out of the labor force**: people who are neither employed nor looking for work (e.g., retirees, students, discouraged workers)
- **Participation rate**: the ratio of the labor force to the total working-age population

Why Do Economists Care about Unemployment?

- Unemployment represents wasted productive resources - people who want to work but cannot find a job
- It directly affects the welfare of the unemployed: loss of income, psychological distress
- It signals the overall state of the economy: high unemployment = weak economy
- A very low unemployment rate can also be a concern - it may signal an overheating economy and rising inflation

Okun's Law
There is a reliable negative relationship between GDP growth and changes in the unemployment rate:
High output growth -> decreasing unemployment
Low/negative output growth -> increasing unemployment

2-3 The Inflation Rate

The GDP Deflator

- **Inflation**: a sustained rise in the general level of prices (the price level)
- **Inflation rate**: the rate at which the price level increases
- **Deflation**: a sustained decline in the price level (negative inflation)
- The **GDP deflator** is defined as:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}}$$

- It is an index number: equal to 1 (or 100) in the base year
- The GDP deflator gives the average price of output - the final goods produced in the economy

The Consumer Price Index (CPI)

- The **CPI** measures the cost of living - the cost of a given basket of consumer goods
- The GDP deflator and CPI generally move together but can differ because:

  > The GDP deflator covers all goods produced domestically; the CPI covers goods consumed (including imports)

  > The GDP deflator uses changing weights; the CPI uses a fixed basket

Why Do Economists Care about Inflation?

- If all prices and wages rose proportionally, inflation would be irrelevant
- In reality, inflation matters because:

  > Not all prices and wages rise proportionally - inflation changes relative prices and redistributes income

  > Inflation creates uncertainty about future prices, making planning harder

  > Inflation interacts with the tax system (e.g., capital gains taxes on nominal gains)

- **Very high inflation** (hyperinflation) is extremely disruptive to the economy
- **Deflation** is also problematic: increases the real burden of debt, and interacts with the zero lower bound on interest rates

2-4 Output, Unemployment, and the Inflation Rate: Okun's Law and the Phillips Curve

Okun's Law

- **Okun's Law**: Above-average output growth leads to a decrease in the unemployment rate; below-average growth leads to an increase
- Quantitative relationship (for the U.S.): A 1 percentage point increase in GDP growth above normal leads to about a 0.4 percentage point decrease in unemployment
- This means that to significantly reduce unemployment, you need sustained above-average GDP growth
- Figure 2-3 plots changes in unemployment vs. GDP growth for the U.S. (1960-2014) - the negative relationship is clear

The Phillips Curve

- **The Phillips Curve**: a relationship between unemployment and inflation
- When unemployment is **below** a certain level (the natural rate), inflation tends to **increase**
- When unemployment is **above** the natural rate, inflation tends to **decrease**
- The 'natural rate of unemployment' is the unemployment rate at which inflation remains stable
- This creates a policy trade-off: reducing unemployment below the natural rate causes inflation to rise
- Figure 2-4 plots changes in inflation vs. unemployment for the U.S. - the negative relationship is visible

2-5 The Short Run, the Medium Run, and the Long Run

Macroeconomists think about economic fluctuations at three time horizons:

The Short Run (Year to Year)

- Focus on movements in output, unemployment, and inflation from year to year
- Key assumption: prices are sticky (slow to adjust)
- Demand plays a central role: changes in demand lead to changes in output
- Monetary and fiscal policy have powerful effects on output
- Models used: IS-LM framework (Chapters 3-6)

The Medium Run (About a Decade)

- Over the medium run, the economy tends to return to a 'normal' level of output
- Prices adjust, and the economy returns to the natural rate of unemployment
- Key factors: labor market institutions, technology, demographics
- Models used: IS-LM-PC model (Chapters 7-9)

The Long Run (Decades to Centuries)

- Focus on what determines the growth of output over long periods
- Key factors: capital accumulation, technological progress, education
- Models used: Growth models (Chapters 10-13)

2-6 A Tour of the Book

The book is organized in four main parts:

The Core (Chapters 3-9)

- Chapters 3-6: The short run (goods market, financial markets, IS-LM model)
- Chapters 7-9: The medium run (labor market, Phillips curve, IS-LM-PC model)

Extensions

- Chapters 14-16: Expectations and their role in the economy
- Chapters 17-20: The open economy (trade, exchange rates)

Back to Policy (Chapters 21-24)

- Chapters 21-23: Fiscal and monetary policy analysis

- Chapter 24: Epilogue - the history of macroeconomics

Appendix: The Construction of Real GDP and Chain-Type Indexes

- Real GDP is constructed using **chain-type indexes** (Fisher index)
- Rather than using a fixed base year, chain-type indexes compute growth rates using prices from adjacent years and then chain them together
- This avoids the problem that a fixed base year becomes increasingly unrepresentative over time
- The BEA uses 2009 as the reference year (where real GDP = nominal GDP)
- For any other year, the growth rate of real GDP is computed by averaging growth rates calculated using prices from two consecutive years